Velocity-Specific Strength Recovery After a Second Bout of Eccentric Exercise

Trevor S. Barss, Charlene R.A. Magnus, Nick Clarke, Joel L. Lanovaz, Philip D. Chilibeck, Saija A. Kontulainen, Bart E. Arnold, and Jonathan P. Farthing

College of Kinesiology, University of Saskatchewan, Saskatoon, Saskatchewan, Canada

Abstract

Barss, TS, Magnus, CRA, Clarke, N, Lanovaz, JL, Chilibeck, PD, Kontulainen, SA, Arnold, BE, and Farthing, JP. Velocity-specific strength recovery after a second bout of eccentric exercise. *J Strength Cond Res* 28(2): 339–349, 2014—A bout of eccentric exercise (ECC) has the protective effect of reducing muscle damage during a subsequent bout of ECC known as the "repeated bout effect" (RBE). The purpose of this study was to determine if the RBE is greater when both bouts of ECC are performed using the same vs. different velocity of contraction. Thirty-one right-handed participants were randomly assigned to perform an initial bout of either fast (3.14 $rad \cdot s^{-1}$ [180$° \cdot s^{-1}$]) or slow (0.52 $rad \cdot s^{-1}$ [30$° \cdot s^{-1}$]) maximal isokinetic ECCs of the elbow flexors. Three weeks later, the participants completed another bout of ECC at the same velocity ($n = 16$), or at a different velocity ($n = 15$). Indirect muscle damage markers were measured before, immediately after, and at 24, 48, and 72 hours postexercise. Measures included maximal voluntary isometric contraction (MVC) strength (dynamometer), muscle thickness (MT; ultrasound), delayed onset muscle soreness (DOMS; visual analog scale), biceps and triceps muscle activation amplitude (electromyography), voluntary activation (interpolated twitch), and twitch torque. After the repeated bout, MVC strength recovered faster compared with the same time points after the initial bout for only the same velocity group ($p = 0.017$), with no differences for all the other variables. Irrespective of velocity, MT and DOMS were reduced after the repeated bout compared with that of the initial bout at 24, 48, and 72 hours with a corresponding increase in TT at 72 hours ($p < 0.05$). Faster recovery of isometric strength associated with a repeated bout of ECC was evident when the velocity was matched between bouts, suggesting that specificity effects contribute to the RBE. The current findings support the idea of multiple mechanisms contributing to the RBE.

Key Words repeated bout effect, specificity, muscle lengthening, interpolated twitch, electromyography, DOMS

Address correspondence to Jonathan P. Farthing, jon.farthing@usask.ca.

28(2)/339–349

Journal of Strength and Conditioning Research

Introduction

Eccentric training produces greater hypertrophy, strength, and neuromuscular gains compared with that produced by isometric or concentric training because of the greater force produced during eccentric muscle actions (14,16). A prominent adaptation that is most commonly observed during eccentric exercise (ECC) is the repeated bout effect (RBE). The RBE is a phenomenon whereby a single exercise session of eccentric contractions protects against muscle damage during subsequent ECC bouts (23,27). The second bout of ECC results in decreased indirect indicators of muscle damage compared with the same time point after the initial bout, including decreased muscle soreness and inflammation, and a faster recovery of isometric strength. The exact mechanism of the RBE is yet to be determined. Recent literature insinuates that the RBE occurs through the interactions of various neural, mechanical, and cellular factors that are dependent on the particulars of the ECC bout and the muscle groups involved (23,26).

The magnitude of the RBE is dependent on the magnitude of the initial muscle damage where greater muscle damage at the initial bout leads to a greater RBE (7,30). Previous research indicates that the velocity at which a contraction is performed contributes to the extent of muscle damage (31). Fast-velocity eccentric contractions may produce higher torques per contraction than do slow velocity contractions (14,17), producing greater muscle damage. Paddon-Jones et al. (31) found that after 36 maximal fast (3.14 $rad \cdot s^{-1}$) or slow velocity (0.52 $rad \cdot s^{-1}$) isokinetic eccentric contractions, both groups experienced a similar decrement in strength but the fast velocity group recovered strength significantly faster. In separate studies, Chapman et al. found that fast-velocity (3.67 $rad \cdot s^{-1}$) ECC caused greater muscle damage than did slow velocity (0.52 $rad \cdot s^{-1}$) ECC when matching for the time under tension (4) and for the number of repetitions when 210 eccentric contractions were completed (5). These studies suggest that differences in muscle damage and strength recovery are evident after single ECC sessions are completed at different velocities. Consequently, there may be a difference in the adaptation to a repeated session of ECC when performed at the same or at a different velocity of contraction.

Previous work has examined the differences in the RBE with fast vs. slow contractions. Chapman et al. (6) recently found that a prior bout of slow velocity (0.52 rad$\cdot$s^{-1}) eccentric contractions reduced muscle damage induced by subsequent fast-velocity (3.67 rad$\cdot$s^{-1}) eccentric contractions. This suggests that the RBE occurs for all eccentric contractions regardless of velocity. However, the authors noted that the magnitude of the protective effect appeared to be reduced compared with the results of previous studies in their laboratory (28,29). Barroso et al. (1) found no difference in the protective capacity of fast or slow velocity eccentric contractions where bouts were performed only at the same velocity. However, no study to date has directly examined the specificity of the RBE by comparing groups of participants that completed each bout at the same velocity with those who completed each bout at different velocities. Therefore, the purpose of this study was to determine whether the magnitude of the RBE is affected by a repeated bout of ECC being performed at either the same or at a different velocity of contraction as the initial bout. It was hypothesized that the protective capacity of the RBE would be greater when the repeated bout of ECC was performed at the same velocity as that of the initial bout compared with when it was performed at a different velocity of contraction.

Methods

Experimental Approach to the Problem

After completing baseline measures to indirectly assess muscle damage and performance, the participants were randomly assigned to perform an initial single unilateral exercise bout of either fast (3.14 rad$\cdot$s^{-1} [180°$\cdot$s^{-1}]) or slow (0.52 rad$\cdot$s^{-1} [30°$\cdot$s^{-1}]) isokinetic eccentric contractions of the elbow flexors. Dependent measures indirectly related to muscle damage and function included maximal voluntary isometric contraction (MVC) strength, muscle thickness (MT), delayed onset muscle soreness (DOMS), mean absolute value (MAV) via electromyography (EMG), voluntary activation via interpolated twitch (ITT), and twitch torque (TT). Indirect measures of muscle damage and function were collected before, immediately after, and at 24, 48, and 72 hours post ECC. After these measurements were taken, each participant took a 3-week break and did not perform any training or exercise of the arms during this duration. Each participant then returned to the laboratory and completed a repeated bout of either slow or fast-velocity ECC. A substantial RBE has been shown to occur with approximately 3 weeks between eccentric bouts (29). Half of the participants from each group were randomly assigned to complete the same velocity or a different velocity ECC protocol as the initial bout. With the exception of the velocity of contraction for the group assigned to a different velocity of contraction on the second bout, the same protocol was followed as that for the initial bout including all baseline measures described above. After the repeated damaging eccentric bout was completed, the dependent variables were measured in the same way as for the initial bout, immediately after, and at 24, 48, and 72 hours posttesting. Figure 1 is a simplified illustration of the study design.

The experimental design resulted in 4 groups that would allow comparison between each velocity of contraction (slow-slow: $n = 8$, 4 men, 4 women; fast-fast: $n = 8$, 3 men, 5 women; slow-fast: $n = 7$, 5 men, 3 women; and fast-slow: $n = 8$, 4 men, 4 women). The primary purpose of this study was to determine if the magnitude of the RBE is affected by whether or not the initial and repeated bouts were performed at the same contraction velocity. The study design allowed for the comparison of 2 groups based on velocity specificity. The 4 initial groups were combined into 2 groups that either completed the same velocity (SPECIFIC: $n = 16$; 7 men, 9 women) of contraction at the initial and repeated eccentric bouts, or a different velocity (NONSPECIFIC: $n = 15$; 9 men, 6 women) of contraction at each bout (Table 1).

Subjects

A sample of 31 participants (men: $n = 16$; and women: $n = 15$) between the ages of 19 and 33 years completed the study protocol. The study was approved by the University of Saskatchewan biomedical review board for research in human subjects, and all the subjects gave their written informed consent according to the American College of Sports Medicine Guidelines. Participants' descriptive statistics are shown in Table 1. All the participants were strongly right handed (Waterloo Handedness Questionnaire 10-item version, scale range from −20 to +20) to control for arm strength asymmetries and the effects of limb dominance. The participants

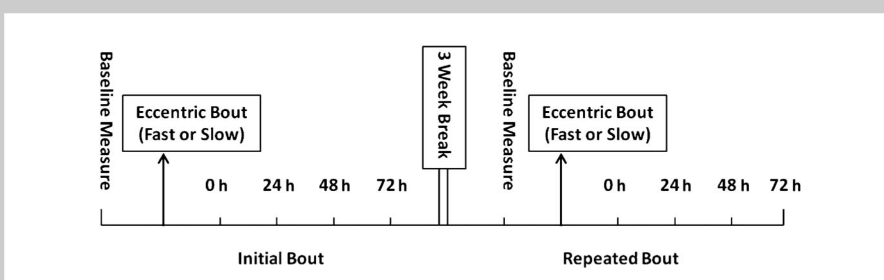

Figure 1. Illustration of the study design. Eccentric exercise bouts were completed on 2 separate occasions separated by 3 weeks. Before, immediately after, and at 24, 48, and 72 hours after each eccentric bout, indirect measures of muscle damage and function were assessed. The participants completed either a fast or a slow velocity of contraction at both the initial and repeated bouts.

Table 1. Subjects' descriptive statistics for SPECIFIC and NONSPECIFIC velocity groups.*†

	SPECIFIC ($N = 16$)	NONSPECIFIC ($N = 15$)
Age (y)	24.8 ± 3.8	23.8 ± 2.1
Height (cm)	174.5 ± 8.6	175.0 ± 8.2
Weight (kg)	77.9 ± 17.3	78.4 ± 14.2
Waterloo handedness score	16.8 ± 2.1	17.6 ± 2.1
Resistance training in previous year (mos)	5.8 ± 4.8	6.5 ± 4.3
Resistance training in lifetime (mos)	42.4 ± 42.5	44.9 ± 34.2
Initial bout total work (N·m)	3,095 ± 1,244	3,328 ± 1,244
Repeated bout total work (N·m)	3,100 ± 1,211	3,468 ± 1,371

*SPECIFIC = group that completed the same velocity of contraction at the initial and repeated eccentric bouts; NONSPECIFIC = group that completed a different velocity of contraction at each bout.
†Values are given as mean ± *SD*.

were free from neuromuscular disease and were instructed to avoid taking any anti-inflammatory agents or nutritional supplements during the experimental period. Although some controversy remains, studies have shown little differences between men and women with respect to relative strength loss, soreness, or skeletal muscle fatigability in response to maximal eccentric exercise (19). As well, there appears to be no difference in the protective capacity of the RBE between men and women (12). Therefore, both men and women were recruited to better generalize to the population. On average, the participants performed 6.1 ± 4.5 months of resistance training in the previous year and for 43.6 ± 38.1 months in their lifetime. No specific eccentric training or isokinetic dynamometry had been performed in the previous year. The participants were asked to limit any strenuous exercise on their upper body for 1 week before the initial bouts of ECC and for the remainder of the study. Maintenance of any existing cardiovascular training program was encouraged. No new resistance, cardiovascular, or flexibility program was to be undertaken during the time frame of the study.

Procedures

Eccentric Bouts. Each maximal voluntary eccentric bout was performed at a velocity of 0.52 $\text{rad}\cdot\text{s}^{-1}$ ($30^\circ\cdot\text{s}^{-1}$) or 3.14 $\text{rad}\cdot\text{s}^{-1}$ ($180^\circ\cdot\text{s}^{-1}$) through a range of motion of 1.92 rad (110°). The participants were familiarized to the eccentric training protocol on the exercise day by completing two 50% eccentric contractions and 1 maximal eccentric contraction before the damaging eccentric protocol at both the initial and the repeated bouts. Previous familiarization was not possible because both submaximal and limited eccentric contractions have been shown to confer a protective effect on future eccentric bouts (21,30). The eccentric bout was performed with the wrist in a supinated position and included 6 sets of 8 maximal eccentric contractions of the elbow flexors, completed on the Humac NORM isokinetic dynamometer (Computer Sports Medicine Inc., Stoughton, MA, USA). Each repetition was completed consecutively with a standardized 1-minute rest between each set. Elbow flexion was chosen because it is common in the RBE literature, and it is a familiar task that produces consistent and reliable data compared with a less familiar task that involves pronounced learning effects (28,30). This volume of work was chosen to ensure adequate muscle damage occurred to see the full protective capacity of the RBE. Approximately, 50 repetitions have been shown in the literature to be a strong stimulus to produce the RBE (18). The participants were placed in a supine position with their feet up on the bench and their opposite arm across their chest to minimize stabilizer muscle activation. During the exercise, the participant's trunk was stabilized with pelvic and shoulder straps to minimize involvement of other muscle groups. The end range of motion was determined by having the participants place their arm (elbow) in full extension (0°) at the side of their bodies where it was most comfortable. The end range for the eccentric contractions was set 10° short of full extension as the participants were unable to produce the threshold level of force needed by the isokinetic dynamometer during eccentric contractions at the end range of motion. Each participant's initial and repeated bouts of ECCs were matched for total work (Table 1). Total work was calculated by summing up the amount of work done throughout each maximal repetition to calculate the total work for each set and each bout. This ensured that the initial and repeated ECC sessions produced similar stress for the elbow flexors. After some careful pilot testing, it was determined that only a few more maximal repetitions were necessary to match the total work between bouts of fast or slow velocity. Total work was monitored throughout the repeated bout of ECC, and when the total work from the initial bout was reached, exercise was halted. If a participant had not completed the required amount of total work after the 6 sets of 8, they were instructed to complete extra repetitions until the required amount was completed. With this method, there were no differences in the total work between initial and repeated bouts of ECC for the SPECIFIC or NONSPECIFIC velocity groups ($p >$ 0.05). Because the displacement of the handle (i.e., range of motion) was the same for fast and slow velocity conditions, the only difference was the time under tension. Standard verbal encouragement was given for all testing contractions.

Maximal Voluntary Isometric Contraction Strength. Maximal voluntary isometric elbow flexion torque (newton meter) was evaluated on the Humac NORM isokinetic dynamometer to assess changes in the MVC strength after the maximal eccentric bouts. The strength assessment was done in the same supine position as in the eccentric bouts with an elbow joint angle of 90°. The participants completed 3 maximal isometric contractions that were each held for 3 seconds and were separated by 1 minute of rest. The contraction with the highest peak force was used for comparison. The participants were familiarized with the isometric strength task by performing a standardized warm-up set of 8 isometric contractions (3 seconds long) on the dynamometer, each separated by 3 seconds of rest. This began at a very low intensity (~25% of maximal effort) and progressively increased to near maximal effort for the last repetition.

Muscle Thickness. Muscle thickness of the biceps brachii (BB) was measured via B-mode ultrasound (Aloka SSD-500, Tokyo, Japan) to assess acute changes related to the inflammatory responses (e.g., edema) after the ECC bout (14,28). A stringent and reliable land-marking method using overhead transparency film was employed to ensure that identical locations were measured at each time point (14,22). The landmark for the center of the ultrasound probe was placed on the muscle belly of the biceps brachii, approximately two-thirds of the way distally down the arm between the acromion and the olecranon process. The participants placed their arm in a relaxed, supinated position on the table with their arm in a standardized lengthened position during each measurement. This technique has demonstrated to be reliable on previous occasions in our laboratory (14,22). Previous work in our laboratory determined the coefficient of variation for the elbow flexors is 1.8% with a test-retest correlation coefficient of 0.98 (14). Four measurements on each arm with the average of the 2 closest were used as the thickness value. Additional measurements were taken in the event that any 2 of the 4 measurements were not within 1 mm. Muscle thickness was assessed before any strength assessment to avoid any transient hypertrophy associated with maximal muscle contractions.

Muscle Soreness. A self-perceived scale of muscle soreness was used to assess the DOMS for a period of 4 days after each eccentric training bout. A visual analog scale (VAS) of a 100-mm continuous line representing "not sore at all" at 0 mm and "very very sore" at 100 mm was used. The VAS scale has been shown to be a reliable and a valid tool for determining muscle soreness (27). The landmark for the center of the ultrasound probe, as described previously, was used as the landmark for muscle belly DOMS assessment. The participants were asked to report DOMS scores with the BB in the same standardized extended position as in MT. An Algometer was used to apply consistent pressure on the muscle belly for each measurement taken.

Electromyography. Electromyographic data were collected to determine the changes in muscle activation during each MVC. An EMG system (Bagnoli-4, Delsys, Boston, MA, USA) was used with 1 surface electrode placed on the BB muscle to measure agonist activation and a second electrode on the triceps brachii muscle to measure antagonist muscle activation. The electrodes were placed on the muscle belly when the arm was positioned at 90° flexion. Land-marking measurements were recorded after the first testing session to ensure the correct placement at each testing time point. A reference electrode was placed on the patella to serve as a common ground for the EMG signal. Raw EMG signal was collected for each of the MVC contractions.

The EMG main amplifier unit included single differential electrodes with a bandwidth of 20 $\pm$ 5 to 450 $\pm$ 50 Hz, a 12 dB per octave cutoff slope, and a maximum output voltage frequency of $\pm$5 V. The overall amplification or gain per channel was 1 K for each muscle. The system noise was $<$1.2 μV (rms). The electrodes were 2 silver bars (10- $\times$ 1-mm diameter) that were spaced 10 mm apart, and had a common mode rejection ratio of 92 dB. Raw data (volts) were later converted to MAV using Matlab (Version 7.3.0 Mathworks, Natick, MA, USA) to examine changes in signal amplitude. A 0.3-second window of time immediately before the ITT was used to calculate the peak MAV. During the testing sessions, the average MAV from the 4 maximal repetitions was used for comparison. A stringent method for land marking the electrodes was used to ensure the accurate placement of the electrodes for each testing occasion (14,22). The skin surface was shaved and cleaned adequately before placing the electrodes. Resting muscle signals were checked for noise, and where appropriate, the skin was cleaned and shaved again, and the electrodes were repositioned. The quality of the skin contact was checked during low-level muscle contractions.

Voluntary Muscle Activation. Percent voluntary activation was estimated using the ITT technique similar to our previous method (22). A Digitimer Constant Current High Voltage Stimulator (Model DS7AH; Digitimer, Hertfordshire, England) was used to maximally activate the elbow flexors. The cathode electrode was placed over the proximal muscle belly of the biceps brachii, whereas the anode electrode was placed over the distal tendon of the biceps brachii. Before the MVC contractions, and starting with a very low intensity of current, a series of resting control twitches with progressively increasing intensity were delivered to the muscle to determine the current intensity (milliamps) needed to elicit maximum resting TT. After the current intensity was identified, the same current intensity was used for the superimposed twitch delivered during each MVC, and the subsequent postcontraction control twitch. Two 4-pulse trains (at 100 Hz, 50-microsecond pulse duration) were manually triggered, both at the peak of the MVC and approximately 5 seconds after the MVC. The level of activation was calculated as voluntary activation (percentage) = (1 $-$ superimposed twitch/estimated resting twitch) $\times$ 100 (33).

Data Acquisition. Custom software in Labview (Version 8.6) was used to obtain stimulator pulses, torque, and 2 channels of EMG data simultaneously. All the channels were acquired at a sampling rate of 1,000 Hz. A desktop computer equipped with an analog-to-digital (A to D) converter was used to convert the analog signals from each device to digital signals displayed in Labview and recorded onto a disk for later analysis.

Statistical Analyses

The raw scores for the MVC strength, TT, and MT data were normalized to the pretest time point values at both the initial and repeated bouts to control for baseline variability between subjects. This was done by calculating the percent change by subtracting the posteccentric bout score from the

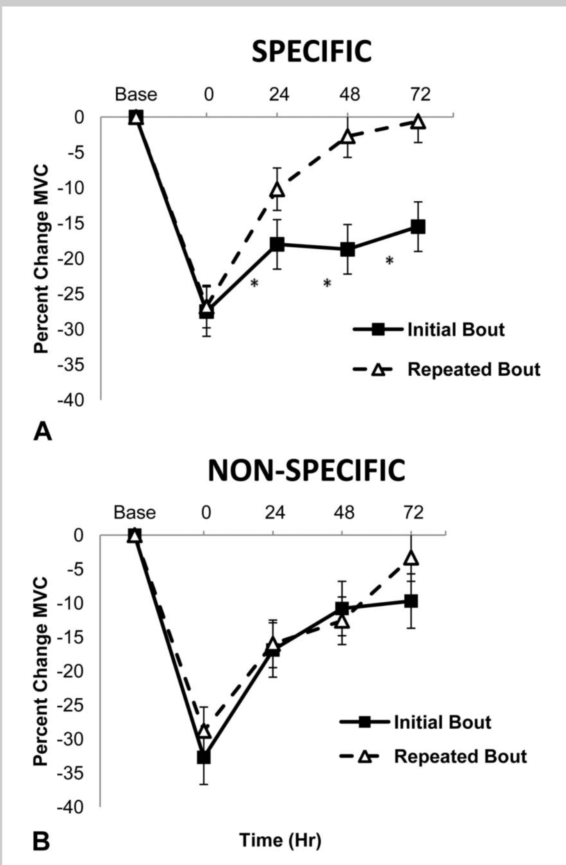

Figure 2. A) Maximal voluntary isometric contraction strength (MVC) percent change (from baseline before the eccentric bout, mean $\pm$ *SE*) for the group that completed the same velocity of contraction at the initial and repeated eccentric bouts (SPECIFIC group). The SPECIFIC group performed the same velocity of contraction at both the initial and repeated bouts of eccentric exercise (ECC; fast-fast or slow-slow). Bout $\times$ Time $\times$ Group interaction ($p = 0.038$). *Initial bout is significantly different from the repeated bout ($p < 0.05$). B) Isometric MVC strength percent change (mean $\pm$ *SE*) for the group that completed a different velocity of contraction at each bout (NONSPECIFIC group). The NONSPECIFIC group performed a different velocity of contraction at the initial and repeated bouts of ECC (fast-slow or slow-fast). There are no significant differences between the initial and repeated bouts for the NONSPECIFIC group.

Table 2. Maximal voluntary isometric contraction strength strength and muscle thickness data for each experimental group.*†

Variable	Group	Bout	Baseline	0 h	24 h	48 h	72 h
MVC strength (N·m)	SPECIFIC ($N = 16$)	Initial	78.6 ± 20.0	57.7 ± 21.1	64.9 ± 20.0	65.0 ± 23.9	67.8 ± 25.7
		Repeated	73.4 ± 24.2	54.3 ± 21.9	65.3 ± 20.1	71.9 ± 26.1§	73.5 ± 27.9
	NONSPECIFIC ($N = 15$)	Initial	80.8 ± 20.5	54.0 ± 18.8	67.0 ± 20.3	72.1 ± 22.8	72.8 ± 20.8
		Repeated	77.3 ± 24.7	55.7 ± 23.0	65.2 ± 21.6	67.7 ± 25.4	74.9 ± 27.3
Muscle thickness (cm)	SPECIFIC ($N = 16$)	Initial	3.90 ± 0.7	4.23 ± 0.7	4.13 ± 0.7	4.09 ± 0.7	4.02 ± 0.7
		Repeated	3.92 ± 0.6	4.15 ± 0.7	4.03 ± 0.6	4.04 ± 0.6	3.96 ± 0.6
	NONSPECIFIC ($N = 15$)	Initial	3.98 ± 0.7	4.25 ± 0.8	4.18 ± 0.8	4.14 ± 0.8	4.16 ± 0.8
		Repeated	3.95 ± 0.7	4.20 ± 0.8	4.06 ± 0.7	4.04 ± 0.7	3.98 ± 0.7

*MVC = maximal voluntary isometric contraction strength; SPECIFIC = group that completed the same velocity of contraction at the initial and repeated eccentric bouts; NONSPECIFIC = group that completed a different velocity of contraction at each bout; ANOVA = analysis of variance.

†Values are given as mean ± *SD*.

‡ANOVA for raw MVC strength revealed a significant bout $\times$ group $\times$ time interaction ($p = 0.042$).

§Initial bout is significantly different from the repeated bout ($p < 0.01$). ANOVA for muscle thickness revealed a bout main effect, where the initial bout is significantly greater than the repeated bout ($p < 0.05$) pooled across group and time.

baseline score (i.e., before the eccentric bout), dividing by the baseline score, and multiplying by 100. In addition, analyses on raw scores were used for MVC strength, MT, DOMS, EMG, and percent activation data.

To meet the primary objective of the study, which was to examine the specificity effects of the RBE, those who completed the same velocity (SPECIFIC) of contraction at the initial and repeated eccentric bouts were compared with those who completed a different velocity (NONSPECIFIC) of contraction at each bout. The analysis consisted of a between-within factorial repeated measures analysis of variance ([ANOVA] [SPECIFIC vs. NONSPECIFIC]), which included 1 between-subject factor of *Group* and 2 within-subject factors of *Bout and Time*. Separate 2 (*Group*; SPECIFIC, NONSPECIFIC) × 2 (*Bout*; initial, repeated) × 4 (*Time*; 0, 24, 48, and 72 hours after the eccentric bout) repeated measures ANOVAs were conducted for MVC strength, TT, and MT data (normalized to baseline before the eccentric bout). Separate 2 (*Group*; SPECIFIC, NONSPECIFIC) × 2 (*Bout*; initial, repeated) × 5 (*Time*; before the ECC bout, and 0, 24, 48, 72 hours after the eccentric bout) repeated measures ANOVAs were conducted for the raw scores for MVC, MT, DOMS, EMG, and percent activation data.

Sphericity violations were adjusted using the Greenhouse-Geisser (GG) method. If significant main effects or interactions were detected, simple main effects analysis continued using 1-way ANOVA and Fisher's least significant difference post hoc tests, or multiple comparisons with adjustment were used where appropriate. Significance was accepted at $p \leq 0.05$. All values are expressed as mean ± *SD*. Analyses were completed using SPSS (version 17, Chicago, IL, USA).

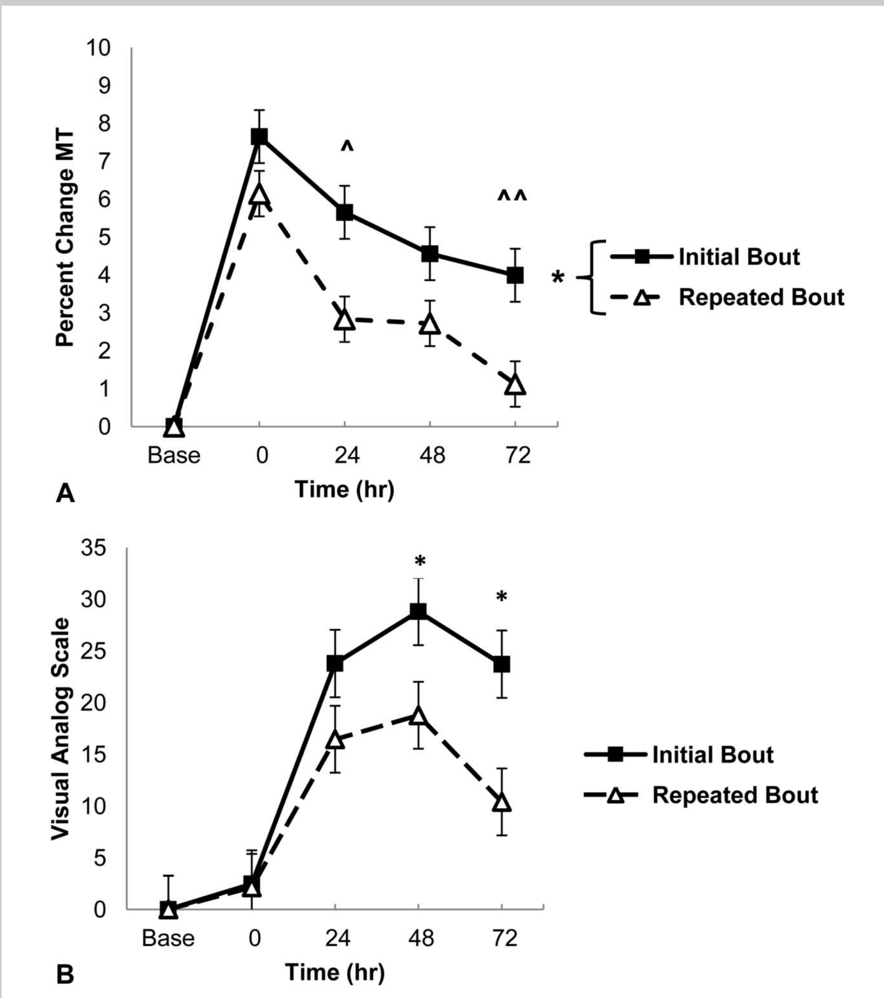

Figure 3. A) Muscle thickness percent change (from baseline before the eccentric bout, mean ± *SE*) pooled across groups. There are no differences between the group that completed the same velocity of contraction at the initial and repeated eccentric bouts (SPECIFIC group) and the group that completed a different velocity of contraction at each bout (NONSPECIFIC group) for muscle thickness percent change. *Significant difference between initial and repeated bout pooled across time ($p < 0.05$). ^Muscle thickness percent change at 24 hours is significantly lower than that at time 0 pooled across bout ($p < 0.05$). ^^Muscle thickness percent change at 72 hours is significantly greater than that at time 48 pooled across bout ($p < 0.05$). B) Muscle soreness (mean ± *SE*) for pooled groups at muscle belly. There are no differences between SPECIFIC and NONSPECIFIC groups for muscle soreness. *Repeated bout is significantly different from the initial bout ($p < 0.05$).

Results

Descriptive Statistics

There were no significant differences between experimental groups for age, height, weight, waterloo handedness score, or resistance training in the past year/lifetime ($p > 0.05$).

Maximal Voluntary Isometric Contraction Strength

The omnibus ANOVA for percent change MVC strength revealed a significant group × bout × time interaction (GG F [2.4, 83.8] = 3.219, $p = 0.038$; Figure 2). Post hoc paired t-tests revealed a significant difference between bouts at time 24 hours (initial bout: −18.0 ± 12.4% vs. repeated bout: −10.2 ± 12.8%; $p = 0.033$), 48 hours (initial bout: −18.7 ± 17.0% vs. repeated bout: −2.7 ± 12.8%; $p = 0.004$), and 72 hours (initial bout: −15.5 ± 18.0% vs. repeated bout: −0.6 ± 17.5%; $p = 0.042$) for the SPECIFIC group. Pooled across bouts, post hoc analysis revealed a significant recovery of strength

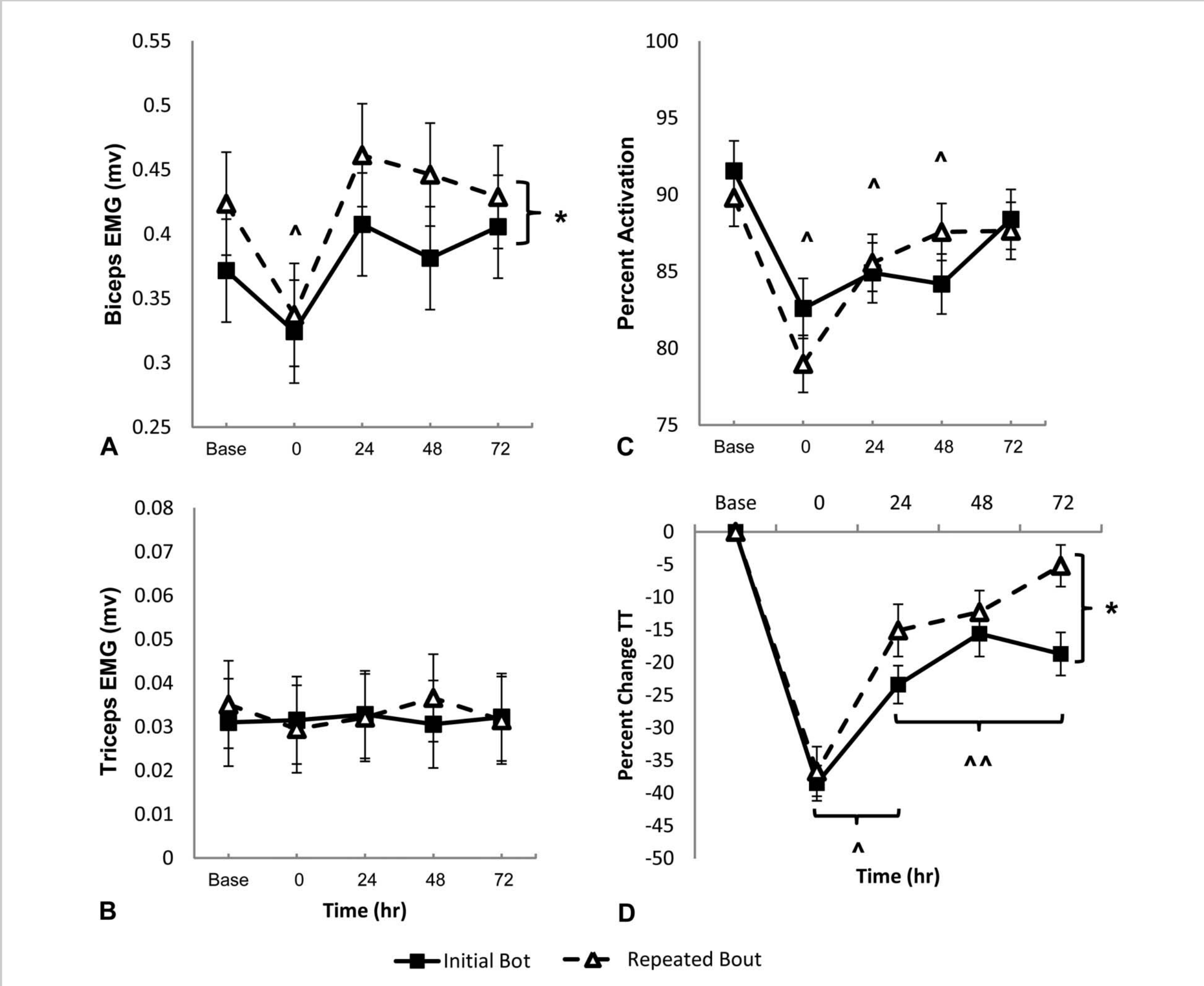

Figure 4. A) Biceps brachii peak mean absolute value (MAV) electromyography (EMG; mean ± *SE*) for pooled groups. There are no differences between the group that completed the same velocity of contraction at the initial and repeated eccentric bouts (SPECIFIC group) and the group that completed a different velocity of contraction at each bout (NONSPECIFIC group) for biceps electromyography (EMG). *Trend for a significant difference between the initial and repeated bouts pooled across time ($p = 0.052$). ^Biceps EMG activation immediately after the eccentric bout is significantly lower compared with that at the baseline pooled across bout ($p < 0.05$). B) Triceps brachii peak EMG (mean ± *SE*) for pooled groups. There are no significant differences for triceps EMG activity. C) Biceps brachii percent activation (mean ± *SE*) via interpolated twitch pooled across groups. ^Biceps brachii percent activation is significantly lower than that of baseline pooled across bouts and groups ($p < 0.05$). D) Biceps brachii twitch torque (TT; mean ± *SE*) pooled across groups. There are no differences between SPECIFIC and NONSPECIFIC groups for TT. *Repeated bout is significantly different from the initial bout ($p < 0.05$). ^The TT is increased significantly from 24 to 48 hours for both the initial and repeated bouts ($p < 0.05$). ^^The TT is increased significantly from 48 to 72 hours for the repeated bout only ($p < 0.05$).

from time 0 to time 24 hours ($p < 0.001$) and from time 48 to time 72 hours ($p = 0.003$). For the NONSPECIFIC group, there was a significant main effect of time ($p < 0.001$) indicating that there was no reduction in strength deficit after the repeated bout. Normalized data are presented in Figure 2 for both groups. Similarly, the ANOVA for raw MVC strength revealed a significant bout × group × time interaction ($p = 0.042$). Post hoc paired *t*-tests revealed a significant difference between bouts at time 48 hours (initial bout: 65.0 ± 23.9 N·m vs. repeated bout: 71.9 ± 26.1 N·m; $p < 0.01$) for the SPECIFIC group. For the NONSPECIFIC group, there was a significant main effect of time ($p < 0.001$). Data for MVC are presented in Table 2.

Muscle Thickness

The omnibus ANOVA for percent change in MT revealed a significant bout main effect (GG $F[1.0, 29.0] = 7.051$, $p = 0.013$; Figure 2A). Pooled across group and time, there was a reduction in the MT after the repeated bout of ECC. However, there was no difference between the SPECIFIC and NONSPECIFIC velocity groups for MT. There was also a significant time main effect ($p < 0.001$). Pooled across bouts and groups, post hoc analysis revealed a significant reduction in the MT from time 0 to time 24 hours ($p = 0.001$) and from time 48 to time 72 hours ($p = 0.027$). Similarly, ANOVA for the MT raw scores revealed a significant bout main effect ($p < 0.05$) with no difference between the SPECIFIC and NONSPECIFIC velocity groups. Data for MT are presented in Table 2.

Muscle Soreness

The omnibus ANOVA revealed a significant bout × time interaction (GG $F[2.3, 67.0] = 4.782$, $p = 0.008$; Figure 2B). There was no difference between the SPECIFIC and NONSPECIFIC velocity groups for DOMS. Post hoc paired t-tests revealed a significant difference between bouts at time 48 hours (initial bout: 28.8 ± 17.4 mm to repeated bout: 18.8 ± 19.8 mm; $p = 0.004$), and 72 hours (initial bout: 23.7 ± 20.3 mm to repeated bout: 10.8 ± 16.1 mm; $p = 0.009$). When splitting by bouts, post hoc analysis revealed a significant increase in DOMS from the baseline to time 0, 24, 48, and 72 hours ($p < 0.05$) for both the initial and repeated bouts, with no difference in DOMS between 24, 48, and 72 hours ($p < 0.05$).

Twitch Torque

The omnibus ANOVA revealed a significant bout × time interaction (GG $F[2.7, 77.0] = 3.086$, $p = 0.038$; Figure 4D). There was no difference between the SPECIFIC and NONSPECIFIC velocity groups for TT. Post hoc paired t-tests revealed a significant difference between bouts at time 72 hours (initial bout: −18.5 ± 18.3% to repeated bout: −5.3 ± 17.8%; $p = 0.009$). Pooled across bouts, post hoc analysis revealed a significant increase in the TT for both the initial ($p < 0.001$) and the repeated bouts ($p < 0.001$) from time 0 to 24 hours. After the repeated bout of ECC only did TT again increase from time 24 to 72 hours ($p = 0.009$).

Electromyography

The omnibus ANOVA for BB EMG activation as shown by peak MAV revealed a trend toward a bout main effect (GG F $[1.0, 29.0] = 4.118$, $p = 0.052$), where the activation was higher after the repeated bout compared with that of the initial bout (Figure 4A). There was no difference between the SPECIFIC and NONSPECIFIC velocity groups. There was also a significant time main effect ($p = 0.004$). Pooled across bouts and groups, post hoc analysis revealed a significant decrease in the EMG activation from baseline to time 0 ($p = 0.005$). The BB EMG activation was no longer different from that at baseline at time 24, 48, or 72 hours. For triceps EMG, there were no significant interactions or main effects (Figure 4B).

Voluntary Muscle Activation

The omnibus ANOVA revealed a significant time main effect (GG $F[3.1, 89.2] = 11.025$, $p < 0.001$; Figure 4C) for percent activation. Pooled across bouts and groups, post hoc analysis revealed a significant decrease in the percent activation from baseline at time 0, 24, and 48 hours, ($p < 0.05$) but no difference compared with the baseline at time 72 hours ($p = 0.411$).

Discussion

The unique finding of this investigation is that the protective capacity of the RBE for elbow flexion MVC strength is greater when the velocity of contraction performed is specific to the initial bout. When a repeated bout of ECC is completed at the same velocity as the initial bout, there is a greater protective capacity for strength recovery than when the bouts are completed at different velocities, supporting our hypothesis. The results also confirm much of those in the previous literature where indicators of muscle damage are reduced after a second bout of ECC, regardless of the velocity of contractions performed (23,26). To our knowledge, this is the first study to demonstrate velocity specificity with the RBE.

We determined a need for this study because previous work has examined the differences in the RBE with fast vs. slow contractions, but no studies have examined the effect of contraction velocity specificity on the RBE. Chapman et al. (6) found that a slow velocity (0.52 $rad \cdot s^{-1}$) bout of ECC reduced muscle damage induced by a second bout of fast-velocity ECC (3.67 $rad \cdot s^{-1}$), suggesting that slow velocity contractions might provide a protective effect for fast-velocity contractions. The current results confirm these findings as inflammation and soreness were reduced after a second bout of ECC, regardless of the velocity of contraction (Figure 3). Barroso et al. (1) found no difference in the protective capacity of fast or slow velocity eccentric contractions of the elbow flexors when the repeated bout was performed at the same velocity of contraction. These results were also confirmed by the findings that the SPECIFIC group had a faster recovery of strength with reduced inflammation and soreness after the second bout of ECC.

An important consideration when comparing fast and slow velocity contractions is the amount of force exerted during each contraction and the time under tension. In this study, both the initial and repeated bouts were completed under the same conditions with the exception of the velocity of contraction, and hence, the time under tension was not matched. To ensure that a consistent amount of force was being placed on the elbow flexors at each bout, the total work for the initial and repeated bouts was matched to allow for standardized comparison. A similar control has been recently used by Barroso et al. (1). To better represent a more realistic training bout, it was decided that 6 sets of 8 repetitions would be completed for each bout regardless of the velocity. Previous studies have chosen to match for time under tension. However, when comparing vastly different velocities of contraction as in this study (0.52–3.14 $rad \cdot s^{-1}$), it would mean completing several times the number of fast-velocity vs. slow velocity contractions. Chapman et al. (4) used this approach to assess the differences in muscle damage with different velocities of ECC and found that the greater number of fast-velocity eccentric contractions produced more muscle damage when matched for time under tension. In the case of this study, 48 slow velocity contractions would be compared with 288 fast-velocity contractions to match for time under tension. From a resistance training applicability standpoint, we considered this an unrealistic comparison that would limit the scope of the results. There is generally support that fast-velocity eccentric contractions produce higher forces than do slow velocity eccentric contractions (14,17). Although conflicting evidence has been presented (3), because total work is the force over a given distance, there may potentially be more force placed on the elbow flexors in the fast compared with the slow velocity bout. Therefore, we chose to control for the total amount of work being performed between the initial and repeated bouts to ensure that any specificity effects associated with the RBE were because of the inherent differences between contraction velocities.

It is well established that there are multiple underlying mechanisms contributing to the RBE (23,26). Reviews by McHugh (23) and Nosaka and Aoki (26) describe evidence for cellular, mechanical, and neural adaptations associated with the RBE. However, specificity effects are generally regarded as a neural adaptation (2,15). Compared with isometric and concentric contractions, eccentric contractions require unique activation strategies by the nervous system where a reduction in muscle activation and an altered recruitment order of motor units may be part of a neural strategy to preserve the highest threshold motor units during eccentric contractions (11). A single ECC bout has been shown to increase BB motor unit synchronization up to 7 days (8,9). As well, motor unit recruitment thresholds are reduced 24 hours after the eccentric bout (10). If smaller changes in motor unit activity or recruitment patterns occur after a second bout of ECC, this could account for the increase in force production during maximal contractions. Because all motor output has both central and peripheral contributions, evidence of specificity of velocity for MVC strength for the SPECIFIC group with no differences for peripheral measures of muscle function and damage (i.e., MT, DOMS, and TT) between groups may provide evidence of adaptation in spinal afferent or central nervous system pathways. The methods used in this study do not allow us to determine the loci of adaptation. However, a shift in the motor unit recruitment strategy may be responsible for the faster recovery of strength in the group that completed the SPECIFIC velocity of contraction at each bout.

After the repeated bout of ECC, there was a significant reduction in MT and DOMS values with a corresponding increase in the TT, which is congruent with the results of previous research. In this study, an RBE was shown for inflammation, muscle soreness, and TT regardless of whether it was in the SPECIFC or NONSPECIFIC velocity group. Twitch torque is a useful measure for muscle contractile properties and excitation-contraction coupling, which can be impaired after damage (36). There was no difference between the SPECIFIC or NONSPECIFIC groups for TT. However, a unique finding was the observation of a significantly greater TT at 72 hours after a repeated bout of ECC indicating an increased ability of the agonist muscle to form crossbridges and produce force (Figure 3D). This result builds on previous ideas that multiple mechanisms contribute to the RBE (23,26). If the RBE was caused entirely by cellular or mechanical mechanisms, a faster recovery of strength would have been seen for both the SPECIFIC and NONSPECIFIC groups regardless of the velocity of contraction. The current results indicate that there is no difference between groups for the amount of inflammation, level of soreness, or ability to produce force via standardized electrical current. Because only those who completed the SPECIFIC velocity at each bout showed a faster recovery for MVC strength, this may indicate that neurologically mediated specificity effects may be contributing to the faster recovery of strength.

The ITT measure was assessed to provide more specific information about a possible neural contribution to the RBE. Twitch interpolation is a peripheral estimation of central activation that provides a valid measure of voluntary activation in muscles (34). Eccentric exercise reduces the ability to voluntarily activate the BB (32). This study confirmed these findings because there was a reduction in the estimated number of motor units being recruited (Figure 4C), up to 48 hours after ECC. However, no differences existed between the initial and repeated bouts confirming the results of Kamandulis et al.'s study (20), who concluded that a repeated bout of ECC of the knee extensors results in reduced muscle damage, but did not influence the level of voluntary activation. The data presented here indicate that there is little contribution of changes in voluntary activation to the RBE as assessed by the ITT.

There was some indication from this study that an increase in neural drive to the agonist muscle might be associated with the RBE. When the participant groups were pooled together, there was a trend ($p = 0.052$) for a significant increase in BB EMG activation after the repeated bout (Figure 4A). The literature has been inconsistent with regard to changes in EMG activation and the RBE. McHugh et al. (24) found that the EMG per unit torque and median frequency were not different between the initial and repeated bouts of isokinetic eccentric knee flexor exercise. They concluded that there was no evidence that the RBE was because of neural adaptation. Falvo et al. (13) found no difference in the median frequency of EMG for a bench press exercise in resistance trained men between initial and repeated bouts. However, in both these studies, submaximal eccentric contractions were used to observe the RBE. It may be possible that the protocol intensity used in these studies was insufficient to demonstrate a change in EMG. Using maximal ECC contractions, Warren et al. (35) showed a reduction in the EMG median frequency in the tibialis anterior after the repeated bout of ECC. Their conclusion was that the torque production was maintained through a greater contribution of slower motor units. Howatson et al. (18) also found a reduction in the EMG median frequency in the BB after a repeated bout of maximal ECC. Most recently, Dartnall et al. (9) showed increases in motor unit synchronization 7 days after an initial maximal eccentric bout of exercise. This is the first evidence of changes in motor unit activation during the repair process after exercise-induced muscle damage. Twenty-four hours after the repeated bout of ECC, they found divergent changes in motor unit synchronization and recruitment threshold compared with the same time point after an initial bout of ECC. This could indicate a neuromuscular adaptation to the initial eccentric bout, contributing to the faster recovery of strength after the repeated bout. The results of this study are not conclusive with regard to the changes in the EMG activity after a repeated bout of ECC. However, a trend for an increase in BB EMG activation after the repeated bout of maximal ECC could indicate an altered motor unit recruitment strategy after the repeated bout of ECC, contributing to the RBE.

There are several limitations to the current investigation. With no direct measure of inflammation, or internal muscle biochemistry, we can only speculate on muscle mechanisms after the eccentric bouts of exercise via measures of TT, MT, and DOMS. Without the use of spinal reflexes or transcranial magnetic stimulation, we are unable to comment on specific sites of neural adaptation using our EMG measure. This information would have been useful for accessing the loci of adaptation but was beyond the scope of this investigation.

The unique result of this study is that MVC strength recovered significantly faster in those who completed both bouts of eccentric exercise at the same velocity of contraction. This is the first study to show specificity of contraction velocity in the RBE. This study also confirms much of the findings of the previous literature where a second bout of ECC was associated with reduced indicators of muscle damage. The reduction in muscle soreness and inflammation with a corresponding increase in TT regardless of contraction velocity indicates that the protective effects of prior ECC are not exclusive to the velocity of contraction used previously. However, the faster recovery of strength seen in those who completed both bouts of ECC at the same velocity of contraction provides indirect evidence for a possible neural contribution to the RBE. The results of this study do not allow for specific sites of adaptation to be identified. Future research is needed to explore the role of spinal reflex pathways, motor unit discharge patterns, and neural drive from the central nervous system to better understand the RBE. The current findings of this study support the idea that there are multiple underlying mechanisms of the RBE.

Practical Applications

This study has practical implications in both athletic and rehabilitation settings. Using eccentric training at velocities that are similar to the performance speed for an athlete may allow for faster strength recovery after a competition or event involving repetitive eccentric contractions. This may allow for an increase in training volume during an athletic season. However, muscle damage from eccentric training will prevent optimal performance until the muscle is fully recovered. Although greater muscle damage from the initial bout of ECC will lead to a greater protective effect, the intensity of the exercise should be taken into consideration to avoid injury and severe DOMS that might impede performance of the muscle in the short term. For athlete groups, the adaptations related to the RBE may be more usefully applied during strength or muscle building phases of training rather than closer to competition. Reversing muscle atrophy or preventing sarcopenia in an elderly population could be optimized through submaximal eccentric training at speeds similar to those of activities of daily living. Incorporating submaximal eccentric contractions could reduce the risk of injury and soreness compared with maximal eccentric contractions, while offering greater muscle load with similar cardiovascular stress (25) compared with the same relative concentric contractions. Providing hypertrophic stimuli while maximizing the recovery of strength after an ECC session may be a beneficial strategy to maintain proper musculoskeletal health.

Acknowledgments

The authors would like to thank the participants who volunteered for this study. The authors declare that they have no conflict of interest. The results of this study do not constitute endorsement of the product by the authors or the National Strength and Conditioning Association.

References

1. Barroso, R, Roschel, H, Ugrinowitsch, C, Araujo, R, Nosaka, K, and Tricoli, V. Effect of eccentric contraction velocity on muscle damage in repeated bouts of elbow flexor exercise. *Appl Physiol Nutr Metab* 35: 534–540, 2010.
2. Behm, DG and Sale, DG. Velocity specificity of resistance training. *Sports Med* 15: 374–388, 1993.
3. Chapman, D, Newton, M, and Nosaka, K. Eccentric torque-velocity relationship of the elbow flexors. *Isokinet Exerc Sci* 13: 139–145, 2005.
4. Chapman, D, Newton, M, Sacco, P, and Nosaka, K. Greater muscle damage induced by fast versus slow velocity eccentric exercise. *Int J Sports Med* 27: 591–598, 2006.
5. Chapman, DW, Newton, M, McGuigan, M, and Nosaka, K. Effect of lengthening contraction velocity on muscle damage of the elbow flexors. *Med Sci Sports Exerc* 40: 926–933, 2008.
6. Chapman, DW, Newton, MJ, McGuigan, MR, and Nosaka, K. Effect of slow-velocity lengthening contractions on muscle damage induced by fast-velocity lengthening contractions. *J Strength Cond Res* 25: 211–219, 2011.
7. Chen, TC, Nosaka, K, and Sacco, P. Intensity of eccentric exercise, shift of optimum angle, and magnitude of repeated bout effect. *J Appl Physiol* 102: 992–999, 2007.
8. Dartnall, TJ, Nordstrom, MA, and Semmler, JG. Motor unit synchronization is increased in biceps brachii after exercise-induced damage to elbow flexor muscles. *J Neurophysiol* 99: 1008–1019, 2008.

9. Dartnall, TJ, Nordstrom, MA, and Semmler, JG. Adaptations in biceps brachii motor unit activity after repeated bouts of eccentric exercise in elbow flexor muscles. *J Neurophysiol* 105: 1225–1235, 2011.
10. Dartnall, TJ, Rogasch, NC, Nordstrom, MA, and Semmler, JG. Eccentric muscle damage has variable effects on motor unit recruitment thresholds and discharge patterns in elbow flexor muscles. *J Neurophysiol* 102: 413–423, 2009.
11. Enoka, RM. Eccentric contractions require unique activation strategies by the nervous system. *J Appl Physiol* 81: 2339–2346, 1996.
12. Eston, RG, Lemmey, AB, McHugh, P, Byrne, C, and Walsh, SE. Effect of stride length on symptoms of exercise-induced muscle damage during a repeated bout of downhill running. *Scand J Med Sci Sports* 10: 199–204, 2000.
13. Falvo, MJ, Schilling, BK, Bloomer, RJ, and Smith, WA. Repeated bout effect is absent in resistance trained men: An electromyographic analysis. *J Electromyogr Kinesiol* 19: 529–535, 2008.
14. Farthing, JP and Chilibeck, PD. The effects of eccentric and concentric training at different velocities on muscle hypertrophy. *Eur J Appl Physiol* 89: 578–586, 2003.
15. Folland, JP and Williams, AG. The adaptations to strength training: Morphological and neurological contributions to increased strength. *Sports Med* 37: 145–168, 2007.
16. Hortobágyi, T, Hill, JP, Houmard, JA, Fraser, DD, Lambert, NJ, and Israel, RG. Adaptive responses to muscle lengthening and shortening in humans. *J Appl Physiol* 80: 765–772, 1996.
17. Hortobágyi, T and Katch, FI. Eccentric and concentric torque velocity relationships during arm flexion and extension: Influence of strength level. *Eur J Appl Physiol Occup Physiol* 60: 395–401, 1990.
18. Howatson, G, VanSomeren, K, and Hortobágyi, T. Repeated bout effect after maximal eccentric exercise. *Int J Sports Med* 28: 557–563, 2007.
19. Hubal, MJ, Rubinstein, SR, and Clarkson, PM. Muscle function in men and women during maximal eccentric exercise. *J Strength Cond Res* 22: 1332–1338, 2008.
20. Kamandulis, S, Skurvydas, A, Brazaitis, M, Skikas, L, and Duchateau, J. The repeated bout effect of eccentric exercise is not associated with changes in voluntary activation. *Eur J Appl Physiol* 108: 1065–1074, 2010.
21. Lavender, AP and Nosaka, K. A light load eccentric exercise confers protection against a subsequent bout of more demanding eccentric exercise. *J Sci Med Sport* 11: 291–298, 2008.
22. Magnus, CR, Barss, TS, Lanovaz, JL, and Farthing, JP. Effects of cross-education on the muscle after a period of unilateral limb immobilization using a shoulder sling and swathe. *J Appl Physiol* 109: 1887–1894, 2010.
23. McHugh, MP. Recent advances in the understanding of the repeated bout effect: The protective effect against muscle damage from a single bout of eccentric exercise. *Scand J Med Sci Sports* 13: 88–97, 2003.
24. McHugh, MP, Connolly, DA, Eston, RG, Gartman, EJ, and Gleim, GW. Electromyographic analysis of repeated bouts of eccentric exercise. *J Sport Sci* 19: 163–170, 2001.
25. Meyer, K, Steiner, R, Lastayo, P, Lippuner, K, Allemann, Y, Eberli, F, Schmid, J, Saner, H, and Hoppeler, H. Eccentric exercise in coronary patients: Central hemodynamic and metabolic responses. *Med Sci Sports Exerc* 35: 1076–1082, 2003.
26. Nosaka, K and Aoki, MS. Repeated bout effect: Research update and future perspective. *Braz J Biomotricity* 5: 5–15, 2011.
27. Nosaka, K and Clarkson, PM. Muscle damage following repeated bouts of high force eccentric exercise. *Med Sci Sports Exerc* 27: 1263–1269, 1995.
28. Nosaka, K and Clarkson, PM. Changes in indicators of inflammation after eccentric exercise of the elbow flexors. *Med Sci Sports Exerc* 28: 953–961, 1996.
29. Nosaka, K, Newton, MJ, and Sacco, P. Attenuation of protective effect against eccentric exercise-induced muscle damage. *Can J Appl Physiol* 5: 529–542, 2005.
30. Nosaka, K, Sakamoto, K, Newton, M, and Sacco, P. The repeated bout effect of reduced-load eccentric exercise on elbow flexor muscle damage. *Eur J Appl Physiol* 85: 34–40, 2001.
31. Paddon-Jones, D, Keech, A, Lonergan, A, and Abernethy, P. Differential expression of muscle damage in humans following acute fast and slow velocity eccentric exercise. *J Sport Sci Med* 8: 255–263, 2005.
32. Prasartwuth, O, Taylor, JL, and Gandevia, SC. Maximal force, voluntary activation and muscle soreness after eccentric damage to human elbow flexor muscles. *J Physiol* 567: 337–348, 2005.
33. Shield, A and Zhou, S. Assessing voluntary muscle activation with the twitch interpolation technique. *J Sport Med* 34: 253–267, 2004.
34. Taylor, JL. Point: The interpolated twitch does/does not provide a valid measure of the voluntary activation of muscle. *J Appl Physiol* 107: 354–355, 2009.
35. Warren, GL, Hermann, KM, Ingalls, CP, Masselli, MR, and Armstrong, RB. Decreased EMG median frequency during a second bout of eccentric contractions. *Med Sci Sports Exerc* 32: 820–829, 2000.
36. Warren, GL, Ingalls, CP, Lowe, DA, and Armstrong, RB. Excitation-contraction coupling: Major role in contraction induced muscles injury. *Exerc Sport Sci Rev* 29: 82–87, 2001.